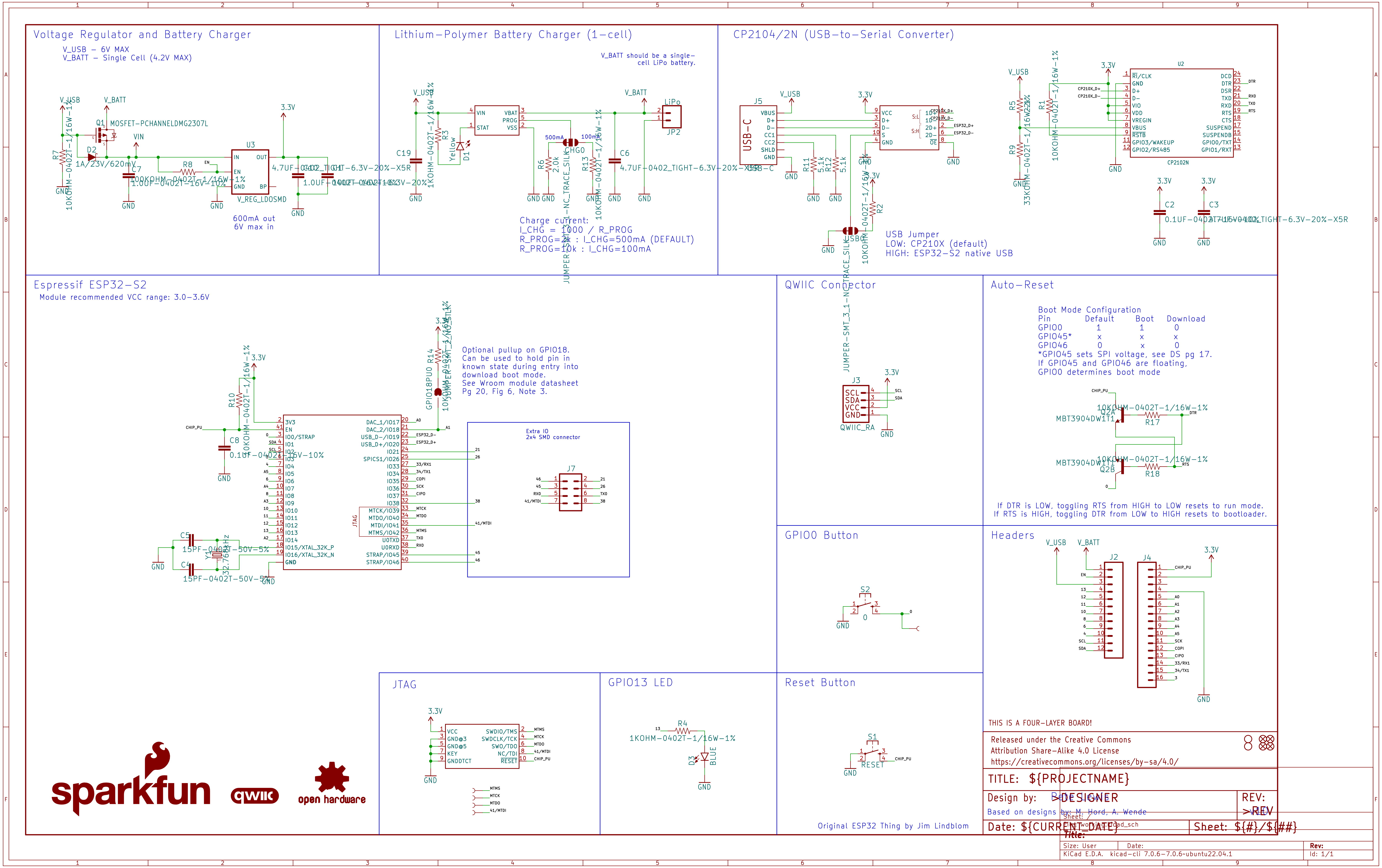

Voltage Regulator and Battery Charger
V_USB - 6V MAX
V_BATT - Single Cell (4.2V MAX)
MOSFET-PCHANNELDMG2307L
V_REG_LDOSMD
600mA out
6V max in
Lithium-Polymer Battery Charger (1-cell)
V_BATT should be a single-cell LiPo battery.
Charge current:
I_CHG = 1000 / R_PROG
R_PROG=2k : I_CHG=500mA (DEFAULT)
R_PROG=10k : I_CHG=100mA
CP2104/2N (USB-to-Serial Converter)
USB Jumper
LOW: CP210X (default)
HIGH: ESP32-S2 native USB
Espressif ESP32-S2
Module recommended VCC range: 3.0-3.6V
Optional pullup on GPIO18.
Can be used to hold pin in known state during entry into download boot mode.
See Wroom module datasheet Pg 20, Fig 6, Note 3.
Extra IO
2x4 SMD connector
QWIIC Connector
Auto-Reset
Boot Mode Configuration
Pin Default Boot Download
GPIO0 1 1 0
GPIO45* x x x
GPIO46 0 x 0
*GPIO45 sets SPI voltage, see DS pg 17.
If GPIO45 and GPIO46 are floating, GPIO0 determines boot mode
If DTR is LOW, toggling RTS from HIGH to LOW resets to run mode.
If RTS is HIGH, toggling DTR from LOW to HIGH resets to bootloader.
GPIO0 Button
Headers
JTAG
GPIO13 LED
Reset Button
THIS IS A FOUR-LAYER BOARD!
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TITLE: ${PROJECTNAME}
Design by: ${DESIGNER}
Based on designs by: M. Hord, A. Wende
Original ESP32 Thing by Jim Lindblom
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
REV:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1
Id: 1/1